

Introduction to Financial Mathematics

This handbook introduces the core calculations used in personal and corporate finance. It is intended as a reference for analysts who need the formulas together with the reasoning behind them.

Chapters cover interest calculations, loan amortisation, present value, and basic risk measures. Each chapter states a formula, defines its variables, explains the underlying mechanism, and works through a numerical example.

Compound Interest

Compound interest is one of the most important concepts in finance. Unlike simple interest, which is calculated only on the original principal, compound interest is calculated on the principal plus all previously accumulated interest. This section introduces the standard formula, defines each variable, explains the mechanism, and works through a numerical example.

The compound interest formula is:

$$A = P(1 + r/n)^{nt}$$

Where the variables are defined as follows:

P = the principal, meaning the initial amount of money invested or borrowed

r = the annual nominal interest rate, expressed as a decimal

n = the number of times interest is compounded per year

t = the total time the money is invested or borrowed, measured in years

A = the final amount accumulated after t years, including all interest

How the formula works. Interest is added to the principal at the end of every compounding period. Once added, that interest itself begins earning interest in each later period. This is the reason growth is exponential rather than linear. Increasing n makes compounding more frequent, which raises the final amount A, although the benefit shows diminishing returns as n becomes large.

Worked example. Suppose $P = 1000$, $r = 0.05$, $n = 12$, and $t = 10$. Substituting gives $A = 1000(1 + 0.05/12)^{(120)}$, which evaluates to approximately 1647.01. The total interest earned over the ten years is therefore about 647.01.

Simple Interest

Simple interest is calculated only on the original principal and ignores any interest already earned.

The simple interest formula is:

$$I = P * r * t$$

Where P is the principal, r is the annual interest rate as a decimal, and t is the time in years. The final amount is $A = P + I$.

Because simple interest never compounds, it grows linearly with time. For the same principal, rate, and duration, simple interest always produces a smaller final amount than compound interest.

Present Value

Present value answers the question: what is a future sum of money worth today?

The present value formula is:

$$PV = FV / (1 + r)^n$$

Where the variables are defined as follows:

PV = the present value, the amount the future sum is worth today

FV = the future value, the nominal amount received later

r = the discount rate per period, expressed as a decimal

n = the number of periods until the money is received

Why discounting is necessary. Money available today can be invested and earn a return, so a rupee received in five years is worth less than a rupee received now. The discount rate r represents the return forgone by waiting. A higher discount rate produces a lower present value.

Risk and Diversification

Diversification reduces the impact of any single asset performing badly. Holding assets whose returns are weakly correlated lowers the variability of the portfolio as a whole without necessarily lowering expected return.

Diversification cannot remove market-wide risk. Systematic risk affects all assets simultaneously and cannot be eliminated by holding more of them.